R2.3 How far? The extent of chemical change

Checklist

What you should know

After studying this subtopic you should be able to:

- State the characteristics of a system at physical and chemical equilibrium.
- Deduce the equilibrium constant equation for a reaction from its balanced chemical equation.
- Determine the composition of an equilibrium mixture from the magnitude of the equilibrium constant, K_c, for a reaction at equilibrium.
- Predict and explain the effect of changes in concentration, pressure and temperature on a reaction at equilibrium.
- Outline how the equilibrium constant, K_c, changes with temperature.
- Explain the effect of a catalyst on the position of equilibrium.

Higher level (HL)

- Calculate the value of the reaction Q for a reaction and determine to which side the reaction will proceed to reach equilibrium.
- Calculate the equilibrium concentrations of reactants and products by constructing an ICE box.
- Calculate the Gibbs energy change from the equilibrium constant, K_c.
- Explain the relationship between the Gibbs energy change, the magnitude of the equilibrium constant and the composition of the reaction mixture for a reaction at equilibrium.